

MINDI TAN

Houston, TX | 832-202-6318 | mindi.tan.w@gmail.com | [linkedin.com/in/minditan](https://www.linkedin.com/in/minditan)

EDUCATION

Texas A&M University | College Station, TX

May 2027

Bachelor of Science in Computer Engineering, Minor in Mathematics

EXPERIENCE

Yugo Apex

March. 2026 – Present

Informational Technology Assistant

College Station, TX

- Supported on-site electricians and maintenance staff with electrical troubleshooting, circuit/wiring checks, hardware inspections, and issue escalation for building systems
- Performed structured troubleshooting on resident-facing technology and building systems, isolating issues across Wi-Fi/connectivity, access control hardware, lighting, circuitry, and wiring while documenting recurring failures and coordinating repairs with on-site staff

TAMU WIRED Robotics - Autonomous Underwater Vehicle

Aug. 2025 - Present

Electrical Team Member

College Station, TX

- Developed and integrated electrical subsystems for an autonomous underwater vehicle, including power distribution, wiring organization, component connectivity, and hardware placement within a compact robot chassis
- Coordinated with mechanical and programming teams to align electrical design with waterproofing constraints, sensor/component placement, motor control needs, and autonomous operation requirements
- Tested and debugged robot hardware during system integration, validating electrical connections, troubleshooting component-level issues, and improving reliability for underwater deployment

NASA Gateway to Blue Skies - Team SOLARA | TAMU ThinkTank

March 2025

Proposal Developer and System Designs Contributor

College Station, TX

- Co-developed SABS, a stratospheric blimp concept to harvest atmospheric water and deliver it to arid farmland; contributed to systems architecture, feasibility analysis, and multidisciplinary proposal documentation
- Authored deployment concepts, system schematics, and trade studies on hydrogen safety, structural durability, and environmental impact to support design decisions under operational constraints
- Developed system design schematics and proposal visuals, including two-blimp architecture, liquid-desiccant water absorption, and hydrogen ventilation analysis ($\sim 8 \text{ m}^3/\text{s}$ airflow) supporting safety and materials trade studies

PROJECTS

Interactive 3D Engineering Portfolio | *Blender, Three.js, Node.js, HTML/CSS, glTF*

Jan. 2026 – Present

- Designed and modeled a custom 3D engineering workspace in Blender, constructing individual scene assets through mesh modeling, object scaling, transform application, UV unwrapping, normal correction, material setup, and hierarchical scene organization
- Prepared production-ready 3D assets for real-time rendering by baking materials/textures, optimizing geometry, reducing scene complexity, configuring lighting/camera views, and exporting assets in glTF format for web integration
- Built and deployed an interactive web-based 3D portfolio using Three.js, Node.js, and HTML/CSS, implementing scene loading, browser rendering, lighting, camera navigation, and interactive visualization of the modeled environment

Sharetea POS System | *Java, JavaFX, PostgreSQL, Git, DAO*

May 2026

- Built a full-stack JavaFX/PostgreSQL POS application for Sharetea with cashier ordering, drink customization, cart checkout, persisted menu data, inventory tracking, and manager-facing operational tools
- Implemented DAO-based database architecture to decouple JavaFX controllers from SQL operations, enabling reliable CRUD functionality for menu items, inventory levels, order records, and transaction history
- Developed reporting and business analytics modules, including sales summaries, restock reports, hourly X-Reports, and end-of-day Z-Reports, while contributing through Git workflows, Agile planning, code reviews, and milestone demos

TECHNICAL SKILLS

Languages: C++, C, Python, Java, HTML/CSS, SQL, MATLAB, Verilog, ARM Assembly

Software/Tools: Git/GitHub, VS Code, Linux, Three.js, React.js, Render

Embedded/Hardware: Xilinx Vivado, I2C/SPI/UART, Digital Logic Design, Sensor Integration, Power Interfaces

Electrical/Lab: Circuit Design & Analysis, Signals & Systems, SPICE/Multism, Oscillosopes, Multimeters, Signal Generators, Logic Analyzers, Soldering, PCB Assembly

3D/CAD: Blender, Three.js, glTF Web Deployment